

CS 171: Problem Set 3

Due Date: October 3rd, 2026 at 8:59pm

1. Two Versions of CPA security (10 points)

There are two common definitions of CPA security, which are given in definitions 0.1 and 0.2 below¹. Prove that definitions 0.1 and 0.2 are equivalent, i.e. if a scheme is secure under one definition, then it is secure under the other definition.

Definition 0.1 Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be an encryption scheme and let $\mathcal{A}$ be an adversary for the CPA security game. Define the CPA security game as follows:

$G_{\mathcal{A}, \Pi}(n)$:

1. The challenger samples a key $k \leftarrow \text{Gen}(1^n)$.
2. The adversary $\mathcal{A}$ is given input 1^n and oracle access to $\text{Enc}(k, \cdot)$, and outputs a pair of messages (m_0, m_1) with $|m_0| = |m_1|$.
3. The challenger samples a bit $b \leftarrow \{0, 1\}$, and computes the ciphertext $c \leftarrow \text{Enc}(k, m_b)$. Then they give c to $\mathcal{A}$.
4. $\mathcal{A}$ continues to have oracle access to $\text{Enc}(k, \cdot)$ and outputs a bit b' .
5. The output of the game is 1 if $b' = b$, and 0 otherwise.

We say that the encryption scheme Π is CPA-secure if for all probabilistic polynomial-time (PPT) adversaries $\mathcal{A}$, there is a negligible function negl such that

$$\Pr [G_{\mathcal{A}, \Pi}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

In definition 0.2 below, any changes from definition 0.1 are shown in red.

Definition 0.2 Let $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be an encryption scheme and let $\mathcal{A}$ be an adversary for the CPA security game. Define the CPA security game as follows:

$H_{\mathcal{A}, \Pi}(n, b)$:

1. The challenger samples a key $k \leftarrow \text{Gen}(1^n)$.
2. The adversary $\mathcal{A}$ is given input 1^n and oracle access to $\text{Enc}(k, \cdot)$, and outputs a pair of messages (m_0, m_1) with $|m_0| = |m_1|$.
3. The challenger computes the ciphertext $c \leftarrow \text{Enc}(k, m_b)$. Then they give c to $\mathcal{A}$.
4. $\mathcal{A}$ continues to have oracle access to $\text{Enc}(k, \cdot)$ and outputs a bit b' .
5. The output of the game is b' .

¹These are analogous to the two definitions of security for EAV security (lecture 3, slides 19-20) and PRGs (lecture 4, slides 8-9)

We say that the encryption scheme Π is CPA-secure if for all probabilistic polynomial-time (PPT) adversaries $\mathcal{A}$, there is a negligible function negl such that

$$\left| \Pr[H_{\mathcal{A},\Pi}(n, 0) = 1] - \Pr[H_{\mathcal{A},\Pi}(n, 1) = 1] \right| \leq \text{negl}(n)$$

2. Feistel Network (10 points)

A Feistel network is used to construct a pseudorandom permutation F given a pseudorandom function f that is not necessarily a permutation². However, if f is not pseudorandom, then F is potentially not pseudorandom either.

Consider the following three-round Feistel network given in definition 0.3 below³.

Definition 0.3 (Three-Round Feistel Network F)

1. Let $f : \{0, 1\}^n \times \{0, 1\}^n \rightarrow \{0, 1\}^n$.
2. **Inputs:** Let F take as input a key $k \in \{0, 1\}^{3n}$ and an input $x \in \{0, 1\}^{2n}$, which are parsed as:

$$k = (k^1, k^2, k^3) \in \{0, 1\}^n \times \{0, 1\}^n \times \{0, 1\}^n$$

$$x = (L_0, R_0) \in \{0, 1\}^n \times \{0, 1\}^n$$

3. Computation:

- (a) F computes $L_1 := R_0$ and $R_1 := L_0 \oplus f(k^1, R_0)$.
- (b) F computes $L_2 := R_1$ and $R_2 := L_1 \oplus f(k^2, R_1)$.
- (c) F computes $L_3 := R_2$ and $R_3 := L_2 \oplus f(k^3, R_2)$.
- (d) F outputs (L_3, R_3) .

Suppose that there was a flaw in the design of f so that for all keys k and all inputs x , the first bit of $f(k, x)$ equals the first bit of x . Show that there exists some efficient adversary $\mathcal{A}$ that can break the pseudorandom permutation security of F by making only a single query to F .

²For more details, see Katz & Lindell, 3rd edition, sections 7.2.2 and 8.6.

³This definition is adapted from Katz & Lindell, 3rd edition, construction 8.23.